

$$\sqrt{0.81}$$

$$0.902$$

$$0.86^{\frac{2}{3}}$$

$$\frac{399}{441}$$

$$= \frac{7x}{(x-5)^2} - \frac{1}{x-5}$$

$$= \frac{7x - (x-5)}{(x-5)^2}$$

$$= \frac{6x+5}{(x-5)^2}$$

Also accept $\frac{7x}{(5-x)^2} + \frac{1}{5-x}$

$$= \frac{7x+5-x}{(5-x)^2}$$

$$= \frac{6x+5}{(5-x)^2}$$

	$(254.9 \times 10^9) \div (2.45 \times 10^6)$	Accept $(254 \times 10^9) \div (2.45 \times 10^6)$ or higher accuracy
	$= 1.04 \times 10^5$	

$$\frac{5c}{2} \div \frac{20c^2}{d}$$

$$= \frac{5c}{2} \times \frac{d}{20c^2}$$

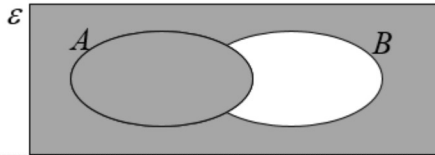
$$= \frac{d}{8c}$$

	$\frac{9kx^2 - kx^2}{kx^2} \times 100\%$	
	$= 800\%$	

	(3-2) units -> \$20	
	Total 9 units -> \$180	

(a)	$-8 \leq 2 - 3x$ and $2 - 3x < 7 - \frac{1}{2}x$	
	$-10 \leq -3x$ and $-5 < 2\frac{1}{2}x$	
	$-2 < x \leq 3\frac{1}{3}$	cannot accept 3.33
(b)	-1, 0, 1, 2, 3	

(a)

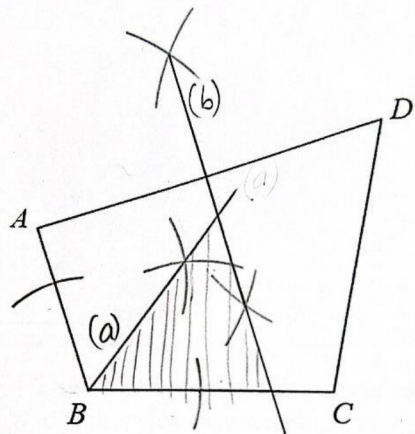


(bi)

$$7 \notin B$$

(bii)

$$\{3, 7\} \subset A$$



Construction arcs are to be clearly seen.

Correct angle bisector

Correct perpendicular bisector

Correct region shaded

$(2p+1)(p-2)=0$	Accept $(2p+1)(3p-6)=0$
$p=-\frac{1}{2}, p=2$	

$$\begin{aligned}\text{Cost price of watch for Jimmy} &= \frac{80}{100} \times 210 \\ &= \$168\end{aligned}$$

$$\begin{aligned}\text{Profit price} &= \frac{120}{100} \times \$168 \\ &= \$201.60\end{aligned}$$

$$\text{Marked price} = \frac{100}{90} \times \$201.60 = \$224$$

(a)	Total time taken $= 15 + 5 + 5 = 25 \text{ min}$ Yes, he will achieve his target as he will complete by 09 15.	
(b)	$\frac{2.3 \times 1000}{25}$ $= 92 \text{ m/min}$	

(a)	$\frac{200}{V} = \left(\frac{1.49}{1}\right)^3$	
	$V = 60\text{g}$	
(b)	Different vertical scales/intervals are used.	

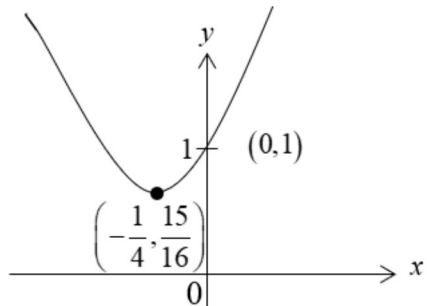
(a)	$1260 = 2^2 \times 3^2 \times 5 \times 7$	
(b)	84 and 180	
(c)	$m = 3, n = 2$	

(a)

$$x^2 + \frac{1}{2}x + 1$$

$$= \left(x + \frac{1}{4}\right)^2 + \frac{15}{16}$$

(b)



(a)	$\mathbf{V} = \begin{pmatrix} 320 & 120 \\ 380 & 100 \\ 410 & 130 \end{pmatrix}$	
(b)	$\mathbf{C} = \begin{pmatrix} 2 \\ 0.5 \end{pmatrix}$	
(c)	$\mathbf{P} = \begin{pmatrix} 700 \\ 810 \\ 885 \end{pmatrix}$	
(d)	$\mathbf{A}$ represents the average ERP charges collected across the three days.	

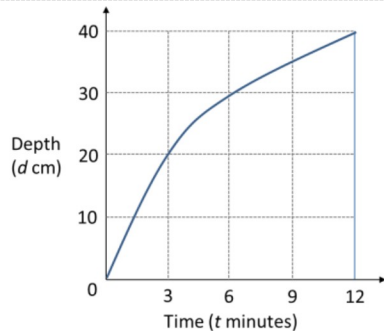
(a)

$$\frac{\frac{1}{2} \times \text{base} \times 20 \times \text{width}}{\frac{1}{2} \times (2 \times \text{base}) \times 40 \times \text{width}} \times 12$$

$$= 3 \text{ min}$$

$$\text{or } \left(\frac{20}{40} \right)^2 \times 12$$

(b)



$$QP = \frac{8}{\tan 0.7 \text{ rad}} = 9.4979$$

$$\text{Area of triangle } OPQ = \frac{1}{2}(8)(9.4979) = 37.992$$

$$\text{Area of sector} = \frac{1}{2}(8^2)\left(\frac{\pi}{2} - 0.7\right) = 27.865$$

$$\text{Area of shaded region} = 10.1 \text{ cm}^2$$

$$(a) \quad \frac{1}{2}(6)v + (18-6)v + \frac{1}{2}(23-18)v = 385$$

$$v = 22 \text{ m/s}$$

$$(b) \quad \frac{30-0}{T-35} = \frac{25-0}{45-35}$$

$$30(10) = 25(T-35)$$

$$T = 47\text{s}$$

$$\text{or } \frac{10}{2} = 2$$

$$\text{or } 45+2=47$$

(a)	angle $PBC = \text{angle } QBR$ (common angle) angle $BQR = \text{angle } BPC$ (corr. angles, $PC \parallel QR$) Triangle PCB and triangle QRB are similar (AA test)	
(b)	Triangles ABC and QRC <i>or</i> Triangles ABQ and CPQ	
(c)	$\frac{CR}{CB} = \frac{3}{7}$	
	$\frac{QR}{PC} = \frac{BR}{BC}$ $\frac{3}{PC} = \frac{4}{7}$	
	$PC = 5.25$	

$$(a) \quad \left(\frac{4a^6}{b^4} \right)^{-\frac{1}{2}}$$

$$= \left(\frac{b^4}{4a^6} \right)^{\frac{1}{2}}$$

$$= \frac{b^2}{2a^3}$$

$$(b) \quad \frac{2^k}{\sqrt[4]{8}} = 4^{2k}$$

$$\frac{2^k}{2^{\frac{3}{4}}} = 2^{4k}$$

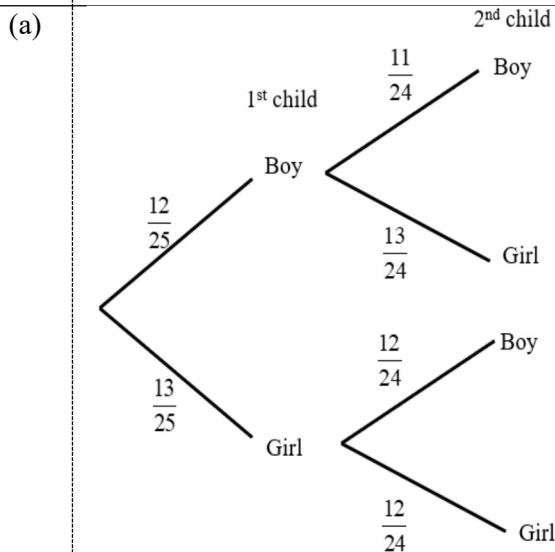
$$k = 4k + \frac{3}{4}$$

$$k = -\frac{1}{4}$$

(a)	$T_n = n(n+1) + 10 - 4(n-1)$ $= n^2 + n + 10 - 4n + 4$ $= n^2 - 3n + 14$	
(b)	$T_{50} = 2364$	
(c)	$n^2 - 3n + 14 = n(n-3) + 14$	
	When n is even $n(n-3)$ is (even x odd) = even. When n is odd, $n(n-3)$ is (odd x even) = even. Adding to 14 which is also even, $T_n = n^2 - 3n + 14$ will always be even for all terms.	

(a)	$\angle EDC = \angle BAE = \frac{(5-2) \times 180}{5} = 108^\circ$	
	$\angle AEB = \frac{180-108}{2} = 36^\circ$	
	$\angle BED = 108 - 36 = 72^\circ$	
	$\angle BED + \angle EDC = 72 + 108 = 180^\circ$	
	By the converse of interior angles, BE is parallel to CD	
(b)	$\angle EBX = 180 - 36 = 144^\circ$	
	$\angle BEC = 108 - 36 - 36 = 36^\circ$	
	As $\angle EBX + \angle BEC = 180^\circ$, by the converse/property of interior angles , $EC \parallel BX$.	Or equivalent methods, with correct reasoning. (eg $BE \parallel CX$ and a pair of opposite equal angles)
	$BE CX$ is a rhombus as $BE \parallel CX$ (or DX) and $EC \parallel BX$ and adjacent sides $BE = EC$.	

(ai)	5	
(aii)	Gradient of PQ = $\frac{-3}{4}$	
	$-2 = \frac{-3}{4}(8) + c$	
	$c = 4$	
	Equation is $y = \frac{-3}{4}x + 4$	
(b)	$2\left(\frac{-3}{4}x + 4\right) - 4x = 19$	Or elimination method correctly follow-thru from a(ii)
	$x = -2, y = 5.5$	
	$(-2, 5.5)$	



(b)

$$\left(\frac{12}{25}\right)\left(\frac{11}{24}\right) = \left(\frac{11}{50}\right)$$

(c)

$$\left(\frac{13}{25}\right)\left(\frac{12}{24}\right) \times 2$$

$$= \left(\frac{13}{25}\right)$$

(d)

$$1 - \left(\frac{12}{25}\right)\left(\frac{11}{24}\right)\left(\frac{10}{23}\right) - \left(\frac{13}{25}\right)\left(\frac{12}{24}\right)\left(\frac{11}{23}\right)$$

or

$$P(\text{BBG}) + P(\text{BGB}) + P(\text{GBB}) + P(\text{GGB}) + P(\text{GBG}) + P(\text{BGG})$$

$$= \left(\frac{12}{25}\right)\left(\frac{11}{24}\right)\left(\frac{13}{23}\right) \times 3 + \left(\frac{13}{25}\right)\left(\frac{12}{24}\right)\left(\frac{12}{23}\right) \times 3$$

$$= 0.78 \text{ or } \left(\frac{39}{50}\right)$$